

Opportunity:

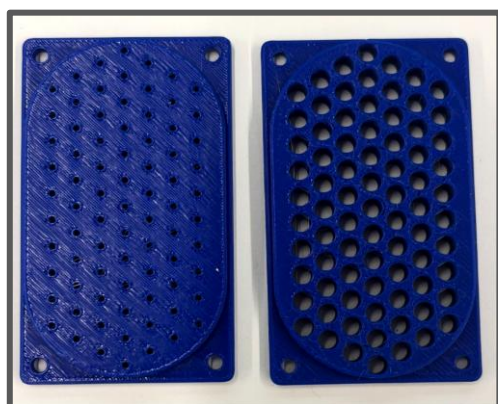
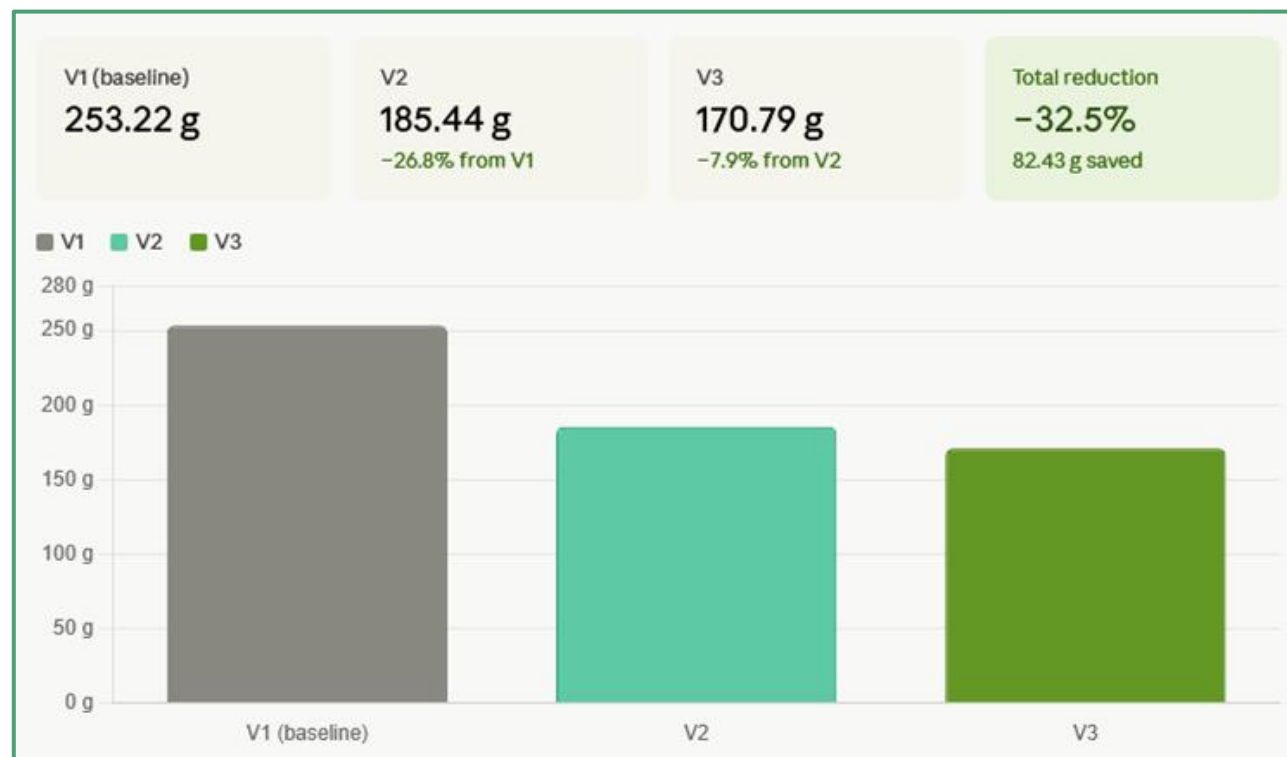
Creating a customizable device that takes physical input and turns it into audio output to improve the communication of non-verbal children.

10 am
Team 5

Key Design Specifications:

- Sound level, dB, ↑
- Battery Life, hrs, ↑
- Portability, g, ↓

Plastic Reduction:



Speaker Grill:

- Tested 6 diameters, + control
- 2.5 mm had loudest output
- Smaller holes muffled output
- Larger holes prone to debris

Word Wizard ✨



Broader Impacts:

- High potential as screen-free learning tool ⇒ larger market

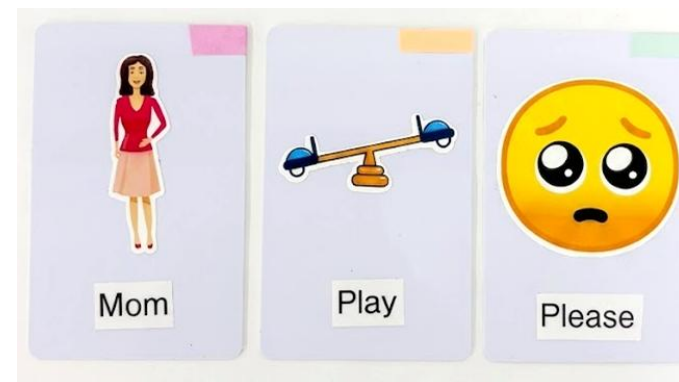
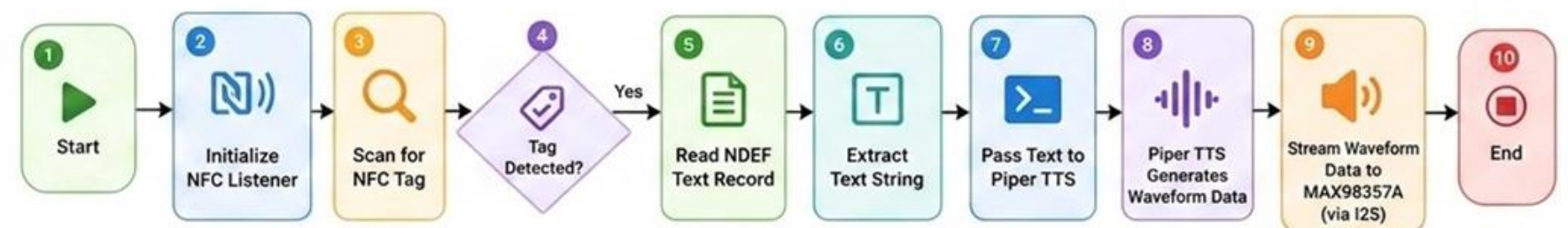
Key Research:

- Minimal mid tech non-verbal communication options exist
- 60-65 dB is needed for high quality sound output

Next Steps:

- Create easier storage system.
- Merge storage with the main device.

Program Flow Chart:

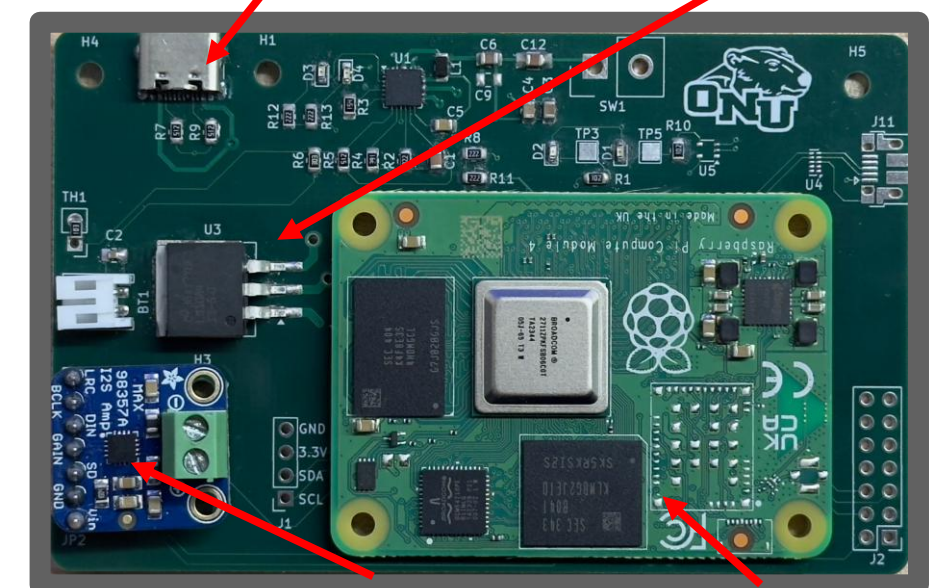


NFC Cards:

- Programmable
- Easy Organization
- 30 Cards

Assembled PCB:

USB-C Charging Voltage Regulator



Speaker Driver Raspberry Pi